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INTERACTION OF HYDROGEN WITH A PALLADIUM (110) SURFACE

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The adsorption of hydrogen on a Pd(110) surface was investigated using Video-LEED, work function ($\Delta\phi$) and thermal desorption mass spectroscopy (TDS) techniques. At 120 K, dissociative adsorption occurs readily with a sticking probability near unity. The formation of a "true" ordered (2×1) hydrogen phase corresponding to $\theta_H = 1.0$ is followed by a (1×2) phase with intense LEED spots indicating a H-induced surface reconstruction. The work function of the clean surface increases continuously with H exposure, up to a saturation value of 325 mV. TDS reveals a total of four hydrogen binding states. Two high-temperature states β_1 and β_2 are produced by chemisorbed H, whereas the low-temperature states α_1 and α_2 only occur on the reconstructed surface and are tentatively attributed to H atoms that have moved to subsurface sites located below the topmost Pd atom layer. The energy splitting of the α states possibly indicates two different kinds of subsurface sites or two different desorption mechanisms; the α_1 site is progressively populated as the surface reconstruction proceeds.

1. Introduction

Owing to its unique properties the Pd/H system has been an interesting candidate for adsorption and permeation studies for many years, and a vast amount of literature exists in this field [1–3]. However, only during the past decade it has been realized that a comprehensive understanding of both the kinetics and the energetics of hydrogen absorption, diffusion, permeation or hydride formation is impossible to attain without the knowledge of the preceding surface processes, namely, of the trapping of the hydrogen molecules from the gas phase and the subsequent dissociation and chemisorption of the H atoms. Studies of these phenomena are therefore essential and are conveniently carried out with Pd low-index single crystal surfaces under ultra-high

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vacuum conditions. There have been reports on the interaction of hydrogen or deuterium with a Pd(100) surface [4] as well as on Pd(111) and Pd(110) surfaces exposed to hydrogen and/or deuterium at room temperature [5,6]. In all these cases, the heat of adsorption of hydrogen was found to range between 20 and 24 kcal/mole, a work function *increase* by 180 to 360 meV was observed, and, except with the (110) surface, adsorption effects could completely be separated from bulk dissolution effects.

Here, we present results of Video–LEED, work function ($\Delta\phi$) and thermal desorption (TDS) measurements obtained with the Pd(110)/H system in the temperature range between 120 and 300 K. Among others, we focussed on the question as to whether surface and bulk phenomena could be delineated as a function of the sample temperature. As will be shown, a quite complex interaction pattern emerges with this crystallographically "open" surface ranging from "simple" hydrogen chemisorption, H-induced surface reconstruction to a selective population of subsurface sites with hydrogen atoms.

2. Experimental

The experiments were performed in a standard stainless steel UHV vessel (Varian) equipped with Video–LEED facilities [7], a cylindrical mirror analyzer for Auger electron spectroscopy, a vibrating Kelvin probe for $\Delta\phi$ measurements, and a quadrupole mass spectrometer (Balzers) for TDS. Further details of the apparatus and procedures used to obtain a chemically clean and defect-free Pd(110) surface can be found elsewhere [8]. As far as the hydrogen adsorption experiments are concerned, we would like to mention that the use of a fast computer-controlled Video–LEED system [7] (data acquisition time about 100 times shorter than with a conventional Faraday cup collector) enabled us to minimize electron beam induced effects on the sensitive hydrogen adlayer (contamination or desorption) during the diffraction experiments. In contrast to previous work from our laboratory [5], all the hydrogen pressures and exposures given in the text are not in N_2 equivalents (gauge reading) but have been corrected with the gauge sensitivity factor for hydrogen of 0.5. The *relative* coverage data (θ_{rel}) have been obtained to within 5% from a numerical integration of the TD curves. The *absolute* coverages (θ_{abs}) cannot be determined with the same accuracy. Here, we have correlated the intensity maximum of the (2 × 1) LEED structure (cf. section 3.1) with the corresponding area of the TD spectrum which unambiguously indicates $\theta_{abs} = 1.0 = 9.35 \times 10^{14}$ H atoms per cm^2, i.e., equal numbers of H and topmost Pd surface atoms.

3. Results and discussion

3.1. Low-energy electron diffraction (LEED)

3.1.1. Adsorption temperature 120 K

The clean and well-annealed Pd(110) surface (which is *not* reconstructed) showed a diffraction pattern with bright and rather sharp spots and negligible background intensity. The hydrogen-induced LEED features are summarized in fig. 1. After an exposure of only 0.1 L at 120 K, faint but sharp "extra" spots of a (2 × 1) pattern appear which go through a sharp intensity maximum at approximately 0.3 L. At the maximum, the fractional-order beam intensity is about 1/20 to 1/50 of that of the integral-order beams. For normal incidence, all beams in the (0, 1/2) positions are missing, which indicates a non-primitive unit mesh containing a glide symmetry plane [9].

We have good reasons to attribute this pattern to a true ordered hydrogen phase on an unreconstructed Pd surface. This evidence comes about from LEED *I–V* measurements of both fractional- and integral-order beams, from absolute intensity considerations, and from the order–disorder behavior to be discussed later. The existence of a glide symmetry plane within the hydrogen (2 × 1) unit mesh demands that it contains at least two hydrogen atoms. This corresponds to a surface coverage of one H atom per Pd surface atom. Since the intensity of the (2 × 1) structure has already reached its saturation after 0.35 L, we deduce a very high sticking probability over this coverage range, similar to recent findings for the Ni(110)/hydrogen system [10].

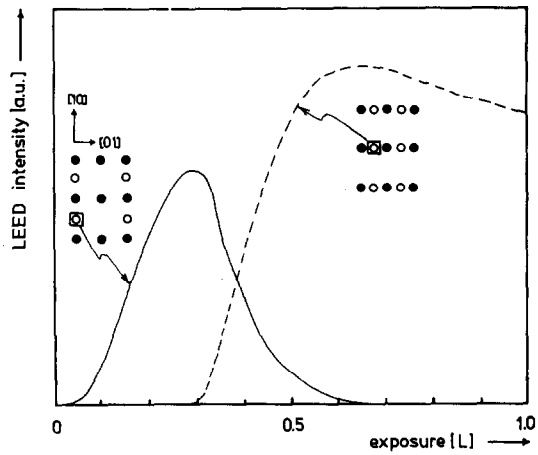


Fig. 1. LEED patterns and intensity–exposure dependences for hydrogen adsorbing on Pd(110) at 120 K. Full line: (2×1) structure, (−1/2, −1) beam, E = 58 eV, 5× enlarged. Dashed line: (1×2) structure, (0, −1/2) beam. Energy: 58 eV.

As soon as the intensity of the (2 × 1) structure has passed its maximum, additional sharp "extra" spots of a (1 × 2) structure develop, which grow in intensity at the expense of the (2 × 1) spots until (after 0.6 L) only the (1 × 2) pattern with *intense* fractional-order beams persists. This high intensity can be taken as a serious hint that the (1 × 2) structure does not reflect a second-order hydrogen phase but rather arises from Pd atoms that have moved to new equilibrium positions under the influence of adsorbed hydrogen atoms. This surface reconstruction might be due to a pairing of adjacent Pd atoms to give the new (1 × 2) periodicity, an assumption that is completely based on analogous arguments for the Ni(110)/H system for which Demuth [11] performed a LEED analysis which favored the pairing model.

Fig. 1 shows that there is an appreciable range in which the two LEED patterns overlap. This indicates that domains or islands of both surface structures coexist. It is probably governed by a nucleation and growth mechanism whereby, wherever the necessary local hydrogen concentration is reached, reconstructed areas form until finally after 0.6 L the entire surface is reconstructed. Since (slight) variations of the temperature do not substantially affect the general pattern depicted in fig. 1, this structural phase transformation requires merely little activation energy and depends only on whether or not a "critical" concentration of surface hydrogen is present. Apparently, a sufficiently large H-induced decohesion of the Pd atoms to which the H atoms are attached must be attained so that they can leave their former equilibrium positions.

The (1 × 2) pattern reaches its maximum intensity after approximately 0.7 L (which, from a comparison with TD spectra, corresponds to roughly $\theta_{abs} = 1.5$). The "extra" spots persist even up to 500 L; however, after about 10–20 L the fractional-order beams begin to get streaky in [01] direction. After about 50 L the sharp (1 × 2) pattern can no longer be observed. Instead, all the "extra" intensity is concentrated in streaks which, at certain electron energies, exhibit distinct spots, e.g., in 1/3 and 2/3 position. Also from the inspection of LEED *I–V* curves of the integral-order beams a further and far-leading degradation of the initial Pd surface topography can be concluded. It is noteworthy in this context that it is just in this range of hydrogen exposures where the TD spectra become more and more dominated by the α states (cf. section 3.2).

3.1.2. Adsorption temperature > 120 K; temperature effects

Fig. 2 summarizes the principal temperature effects on the two hydrogen-induced LEED structures described above and generated at 120 K. If we start heating from a regime (0.4–0.6 L) where both LEED patterns coexist, we obtain the full line for the (2 × 1) and the dashed line for the (1 × 2) beam intensity. Between about 160 K and 170 K there is a sharp rise in the (2 × 1) intensity up to a maximum at 180 K; further heating leads again to a continuous decrease of the intensity because of increasing thermal disorder.

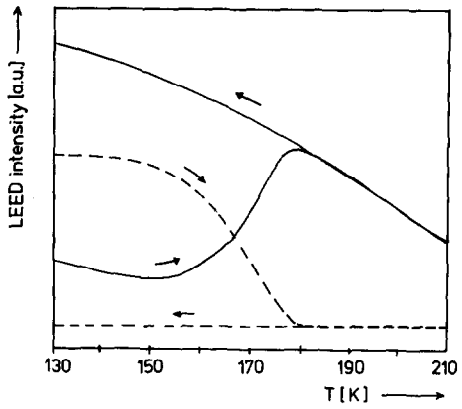


Fig. 2. Temperature dependence of the LEED intensities. Full line: $(-1/2, -1)$ beam of the (2×1) structure after 0.4 L H_2 adsorption at 120 K. Dashed line: (1×2) structure, $(1, 1/2)$ beam, $E = 58$ eV, after 0.4 L H_2 at 120 K. The arrow indicates the direction of the experiment.

Cooling down to 120 K is reversible only down to 180 K; thereafter a pronounced hysteresis occurs insofar as the (2×1) intensity continues to increase: Being back at the starting temperature of 120 K, the (2×1) intensity

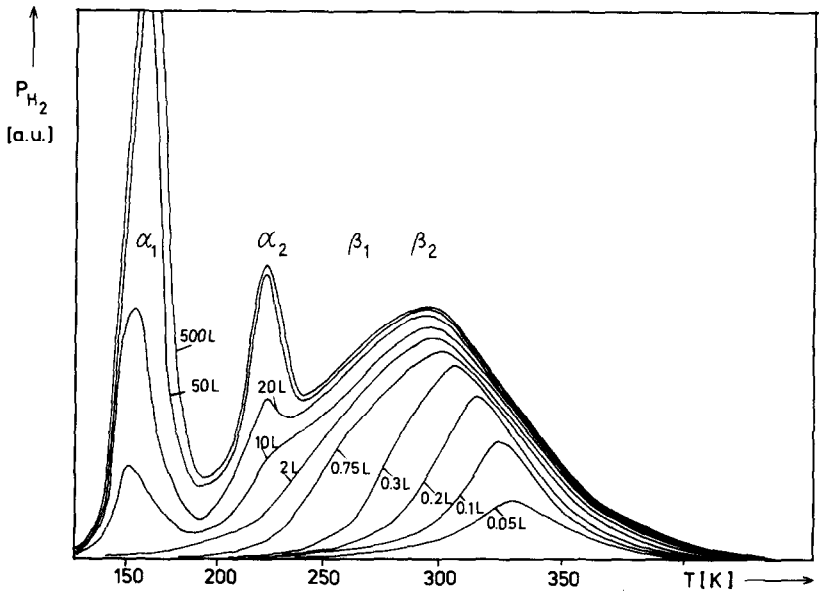


Fig. 3. Series of thermal desorption spectra of H_2/Pd(110). The adsorption was performed for various exposures at $T = 120$ K. The various desorption states are marked in the figure. The heating rate was 6 K/s.

has reached almost the highest possible value – close to the maximum which is reached after 0.3 L at 120 K (cf. fig. 1). The (1 × 2) spots, on the other hand, begin to get weaker as the surface temperature approaches 170 K and they irreversibly lose most of their intensity between 180 and 200 K, depending on the initial coverage. In fig. 2, the (1 × 2) intensity is down to zero already near 185 K. Cooling down again has the surprising result that the (1 × 2) intensity remains zero! These LEED observations could easily be rationalized if the amount of hydrogen required to form the (1 × 2) structure would thermally desorb to just such an extent that the amount associated with the (2 × 1) intensity maximum, i.e., $\theta \approx 1.0$, would always be left on the surface. An inspection of the thermal desorption spectra (cf. fig. 3), however, immediately reveals that heating to 220 K (where in the above experiments the heating was stopped and the cooling started) does not remove enough hydrogen atoms from the surface by thermal desorption. Instead, it is much more likely that the H atoms use a different channel to escape from the surface, namely, they diffuse into the bulk material. The corresponding diffusion paths, however, are quite short; indeed we have evidence that the escaping H atoms are still in contact with the (2 × 1) H phase and form a so-called "subsurface" species: As the reconstructed (1 × 2) surface relaxes into the unreconstructed form, also the Pd atoms return to their former positions and may very well sort of "bury" the H atoms underneath in that the diffusion channels *to* the surface become blocked. We will come back to this model when we discuss the TDS data.

One other aspect deserves attention: If we concentrate on the low coverage regime, i.e., from $\theta_{abs} = 0$ to $\theta = 1.0$, we only deal with the (2 × 1) phase, and we can in principle use the LEED intensity of the fractional-order beams as a direct monitor of the long-range order within the hydrogen adlayer. So we should be able to follow the order–disorder transition as a function of coverage and temperature as was done in our laboratory for the systems H/Ni(111) [12], H/Pd(100) [4], and H/Fe(110) [13] previously. However, with the "open" (110) face of palladium, bulk diffusion of hydrogen which is also quite temperature dependent seriously impairs such order–disorder measurements in that (reversible) equilibrium curves cannot be obtained.

Although no extensive LEED measurements were carried out to determine the influence of the adsorption temperature on the surface structure, some additional observations should be mentioned here. As soon as the adsorption temperature is raised to beyond 210 K, the (2 × 1) structure can no longer exist. Nevertheless appreciable amounts of hydrogen atoms are still adsorbed and form a disordered phase. Adsorption at or near room temperature gives rise to the same streaky (1 × 2) pattern that is observed at 120 K after prolonged exposure of hydrogen, but interestingly the sharp (1 × 2) pattern cannot be obtained under any circumstances, not even by cooling down a streaky pattern generated at room temperature.

3.2. Thermal desorption spectroscopy (TDS)

Fig. 3 displays a series of thermal desorption spectra from a Pd(110) surface exposed to hydrogen at 120 K. The exposure was varied from 0.05 to 500 L. At low exposures, a single fairly broad maximum appears at 325 K, denoted as β_2. It arises from chemisorbed H and shifts down to lower temperatures as the coverage increases pointing to a second-order rate process. It reflects those H atoms that have formed the (2×1) structure below 200 K, although by the time they are desorbing their long-range order has completely broken down. As the exposure is further increased, e.g., to 0.75 L, the (2×1) pattern has been replaced by the reconstructed (1×2) phase, but again, the hydrogen that we see in the TD curve has desorbed from the *unreconstructed* surface because we know from the foregoing LEED section that, no matter how much hydrogen is adsorbed, the (1×2) reconstruction is removed around 180–200 K. The shape of the TD curve suggest that it contains a second hydrogen binding state, β_1, as a shoulder, but a clear separation of β_1 from β_2 is not possible.

As we have pointed out in the foregoing section there is ample evidence that, once the surface is reconstructed, channels open up for hydrogen to diffuse below the surface, particularly when the reconstruction has reached the stage of the "streaky" LEED pattern. This is proven by two additional low temperature states α_1 and α_2 which, like the β states, contain *atomic* hydrogen but are much narrower than those. The α_2 state peaks at 225 K, exhibits an approximate first-order kinetics and saturates after roughly 50 L, whereas the α_1 state desorbs via fractional-order kinetics at $\sim$ 160 K and does not saturate up to a 500 L exposure. However, even after 500 L the area under the α_1 peak is only equal to or less than the area of the β peaks; the area under the α_2 peak is even less. This means that the total amount of hydrogen of the α states is about equal to that of the β states. Also, the sticking probability for adsorption into the α states is quite small. It has to be remembered that under our experimental conditions no high temperature desorption of hydrogen was found. This is in contrast to the results of Conrad, Ertl and Latta in their room temperature study of H adsorption on Pd(110) [5], where they reported on a broad H desorption feature around 600 K which they interpreted as bulk hydrogen leaving the surface in a diffusion-controlled reaction.

The origin of our α states is clearly related to the heavy surface reconstruction occurring at exposures above 10 L, and in the following we present one of various possible explanations. The reconstruction process undoubtedly creates a new active surface, namely, the atoms of the second Pd layer become accessible for adsorption of hydrogen. Furthermore, the H atoms are offered more channels to enter the Pd lattice and, depending on their concentration (which, in turn, is governed by the exposure), have the possibility to accumulate in subsurface or surface-near bulk regions of the crystal. Now, as the surface is heated, the rate of diffusion increases and part of the H atoms reach

the surface again from which they desorb into the α_1 state without "feeling" the deep potential well of the (still occupied) chemisorption sites. Note that these H atoms penetrate through the "open" reconstructed surface. Further heating removes the reconstruction around 180–200 K, and the formerly freely accessible second Pd layer is screened again by the relaxing first layer atoms. As a consequence, the "easy" diffusion exit paths which were allowed for the α_1 H atoms are no longer available. Therefore an additional activation energy is required for the α_2 H atoms to move through the topmost layer and to escape into the gas phase. This process seems to be the bottleneck of this desorption reaction and not the recombination to form the H_2 molecule as is indicated by the first order kinetics. At any rate, our thermal desorption results are strong evidence that indeed a subsurface hydrogen species as proposed recently [14] can exist, but to our opinion an "open" surface and a reconstruction during the adsorption reaction are necessary prerequisites. Additional evidence for our interpretation is provided by the work function results which will be described next.

3.3. *Work function ($\Delta\phi$) measurements*

The variation of the work function of a Pd(110) surface upon exposure to hydrogen at 120 K is shown in fig. 4. Interestingly, a completely continuous curve is obtained: Neither the formation of the ordered (2×1) phase (maximum order after 0.3 L) nor the beginning reconstruction to form the (1×2) phase after 0.5 L can be distinguished in the $\Delta\phi$–exposure relation. After 0.8 L a saturation value of 325 mV is obtained which is in fair agreement with the 360 mV reported for room temperature adsorption previously [5]. We note that

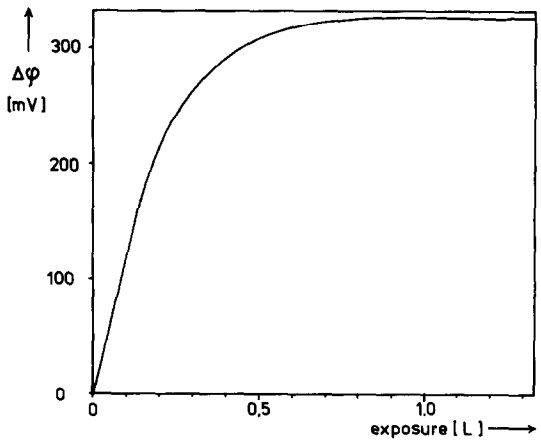


Fig. 4. Variation of the H-induced work function change with exposure at $T = 120$ K.

in TDS, saturation for the β states was reached after 0.8 L exposure. This means that the population of the α states does not contribute to the work function change to a measurable extent. We take this as another hint to the filling of subsurface sites that do not produce a surface dipole moment.

A much more complex pattern is obtained if the surface is charged with hydrogen at 120 K and then linearly heated (3 K/s) to allow monitoring of the variation of the work function change. The resulting curves for various initial H coverages are reproduced in fig. 5. Concentrating on the 0.1 L curve first, which describes the thermal decomposition and desorption of the (2 × 1) phase only, we see that there is a very small $\Delta\phi$ variation, perhaps associated with the order–disorder transition around 210–230 K, and only the thermal desorption of the hydrogen brings the work function back to the clean surface value whereby the point of inflection coincides nicely with the pressure TD maximum of fig. 3. The 0.8 L curve, on the other hand, exhibits similar features but in addition shows another steplike decrease of $\Delta\phi$ in the temperature range between 150 and 190 K, i.e., just in the region where the reconstruction is removed. We therefore associate this work function step of approximately 75 mV with the $\Delta\phi$ between the (1 × 2) phase on the reconstructed surface and the (2 × 1) phase on the unreconstructed surface. There are no further changes when we go to curves with a higher initial exposure: Clearly the surface work function is saturated after 0.8 L and again we have evidence that the α states

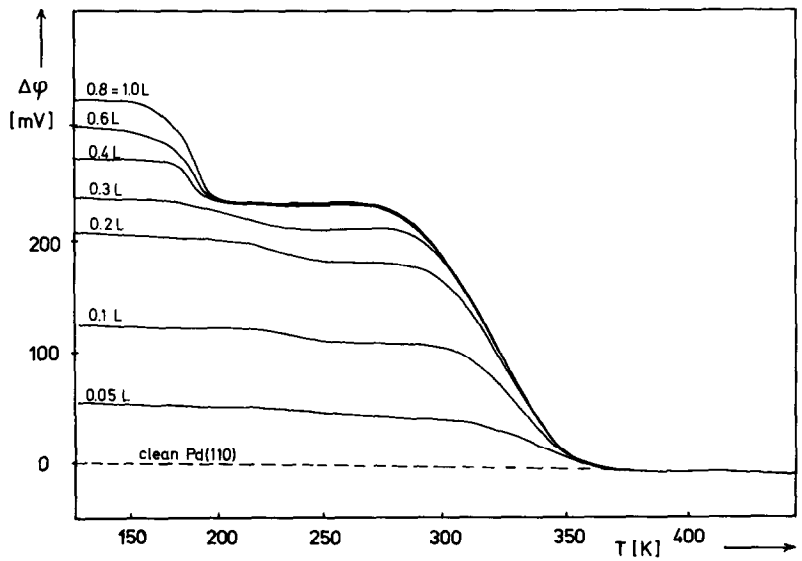


Fig. 5. Temperature dependence of the work function of a Pd(110) surface covered with various amounts of hydrogen at $T = 120$ K. The heating rate was 3 K/s. See text for further details.

do not give rise to a work function change. The irreversibility of the $(1 \times 2) \rightarrow (2 \times 1)$ phase transformation seen in the LEED (cf. section 3.1, fig. 2) is also reflected in these thermal work function measurements: If we heat a surface exposed to 0.6 or 0.8 L hydrogen at 120 K (which contains the (1×2) phase), just to 220 K we do see the initial steplike decrease of ~ 75 mV, but upon cooling down again the work function remains at 250 mV and does not increase anymore.

4. Conclusions

We have shown in the present study that interaction of hydrogen with a Pd(110) surface leads first to a true ordered hydrogen (2×1) phase. Beyond $\theta = 1$ the surface undergoes reconstruction the extent of which strongly increases with exposure. This enables H atoms to enter the near-surface regions of the crystal and to populate subsurface sites from which they desorb again at higher temperatures.

Acknowledgements

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